Sentiment Analysis with a Recurrent Neural Network: Album Rating Prediction

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Abstract

For my project, I will be training an RNN model to predict a user's score of an album based on their review of it. I will train it and test it with data from a music review website, and then see how it compares to guesses from an LLM and guesses from a human.

1. Overall project goal

My project will be training a model to perform sentiment analysis on music reviews. My main goal is for it to be able to predict a user's rating of an album based on their written review. I don't have a specific benchmark in mind at the moment, but I just want to tweak it to work the best that it can by the deadline.

2. Problem description

I will be investigating whether a recurrence neural network can be implemented to accurately predict a user's score of an album based on the review that they wrote about it. I think that it's interesting because the naive approach these days would be to just feed the review into a large language model and then ask it for its prediction, but on a large scale, this would be quite resource-intensive. I'd like to find out for myself how much effort the RNN approach will take, how it performs, and whether the process is worth it in the end.

2.1. References

I will review the "RNNs and LSTMs" chapter of *Speech and Language Processing* to provide context and background for this project. In addition, I will be learning the process of properly training and utilizing the model from TensorFlow's tutorial on "Text classification with an RNN."

3. Data

I will be harvesting my own data by web scraping a review site like Rate Your Music or Album of The Year. In this way, I can read and compile many reviews and ratings from the most popular albums on the site in an automated manner that I can then use to train, validate, and test the accuracy of the model.

4. Approach

I plan to implement a recurrence neural network. I will absolutely be referring to other RNN tutorials, projects, and implementations along the path of development, but I do plan to train one myself rather than simply testing out a preexisting plug and play model to immediately make predictions on my data.

5. Baselines

My baseline method would be TensorFlow's recurrence neural network which can be trained to determine whether the sentiment of a review is positive or negative. I will be building upon this so that it can predict a specific score. Because I'll be training the model for a different application with different data, I'll need to run my own experiment with this method.

6. Evaluation

I would like to evaluate my approach by seeing how it compares to asking an LLM to predict a user's score given their review. I'd also like to see how it compares to a person guessing. However, it is worth noting that in both of these cases, there are resource constraints, being token limits and human participation, respectively, so I may not be able to obtain the most data from these alternate methods.

References

[1] D. Jurafsky, and J. H. Martin, "RNNs and LSTMs," *Speech and Language Processing*, pp. 1–27, Jan. 2026.

[2] TensorFlow, "Basic text classification," Aug. 2024.

[3] TensorFlow, "Text classification with an RNN," Mar. 2024.

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